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Feature-weighted calibrated tabular explainable AI for heart disease prediction

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Abstract

Cardiovascular disease (CVD) remains a leading cause of mortality and morbidity globally. Early detection and stratification of CVD risk using machine-learning models can improve patient outcomes; however, existing models often suffer from limited interpretability, small sample sizes, class imbalance, and uncalibrated probability outputs. This paper proposes FWC-TabXAI++, a feature-weighted calibrated tabular explainable AI framework for heart disease prediction. The method employs clinically constrained data preprocessing, second-order feature-interaction augmentation, mutual-information-based feature weighting, a heterogeneous ensemble of standard methods, validation-informed probability blending, and threshold selection. Its output includes probability estimates assessed for calibration, SHAP values, and LIME-based global and local explanations. The framework was tested on a cleaned clinical dataset comprising 836 records. FWC-TabXAI++ achieved an accuracy of 91.07%, a precision of 92.22%, a recall of 91.21%, a specificity of 90.91%, an F1-score of 91.71%, and an ROC-AUC and PR-AUC of 94.01% and 93.24%, respectively, and achieved the best overall balance across performance metrics. The model also exhibited a low Brier score of 9.08%, and bootstrap confidence intervals supported the internal stability of the proposed framework. SHAP and LIME analyses identified ST-segment slope and exercise-induced angina as dominant predictive contributors and provided patient-centred explanations. These results suggest that FWC-TabXAI++ offers a robust and interpretable approach for CVD risk prediction.

Policy significance statement

This research supports healthcare providers and policymakers by enabling transparent, data-driven heart disease risk assessment using interpretable machine learning. The framework may improve early screening, patient prioritisation, and clinical decision support while maintaining clinician oversight.

Keywords Cardiovascular disease, Ensemble learning, Feature weighting, Explainable AI, Calibrated probability

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1 Introduction

Cardiovascular disease (CVD) is still the most prevalent cause of illness and death around the world. The key to improving outcomes for these patients is early identification and risk stratification [1–4]. Over the last few years, the application of machine learning (ML) in the development of clinical decision support systems for identifying and predicting CVD has advanced considerably. These techniques are deployed to model the complex relationships among demographics, electrocardiograms, and blood biochemistry that would otherwise be undetectable with conventional statistical methods. However, the current limitations of ML-based CVD predictors are many. Some research is built on small or single-centre datasets or uses black-box deep learning models that are not explainable or interpretable [5, 6]. Other research focuses solely on accuracy at the expense of calibration and class imbalance. Although hyperparameter tuning and advanced ensembles have improved prediction accuracy, many existing studies do not jointly incorporate second-order feature interactions, calibrated probability estimation, statistical validation, and global–local explainability. Moreover, only a limited number of studies provide both global and local explanations to support clinical decision-making [7–9]. To address these gaps, this study formulates heart disease prediction as an interpretable, calibrated binary classification problem. The objective is to provide accurate and clinically useful patient-level CVD risk prediction while explaining both global and local feature contributions. The main contributions of this paper are as follows.

- We propose FWC-TabXAI++, a feature-weighted, calibrated ensemble framework that expands second-order feature interactions and applies mutual-information-based feature weighting and selection.
- A heterogeneous collection of baseline models is utilised, including logistic regression, support vector machines, k-nearest neighbours, random forests, and gradient boosting. The optimal model blend is learned from validation data, and the decision threshold is adjusted to maximise the F1 score.
- The bootstrap method is used to construct confidence intervals, and calibrated probabilities and uncertainty quantification are obtained. Global and local interpretations are provided based on SHAP and LIME, respectively.

The proposed method demonstrates effectiveness on a cleaned, stratified heart disease dataset. A wide range of experiments was conducted, including ablation studies, comparisons of different ensembles, and analyses of calibration and statistical validation based on bootstrapping. The remainder of this paper is structured as follows: Sect. 2 reviews recent studies on ML-based heart disease prediction. Section 3 details the data and methods, including data and feature preparation, ensemble modelling steps, and calibration and interpretability methods. Section 4 discusses the results of the experiments. Section 5 provides the conclusion.

2 Literature survey

The literature on cardiovascular disease (CVD) prediction using machine learning has grown rapidly, and researchers have explored diverse classification algorithms, ensemble methods, feature selection strategies and explainable AI to improve prediction accuracy and clinical utility. Mohapatra et al. [10] used a two-level stacking ensemble for heart anomaly detection and CVD prediction. Their ensemble consisted of heterogeneous

base classifiers, including logistic regression, decision trees, k-nearest neighbours, random forests, and SVMs. A meta-level classifier consolidated the base predictions. Their stacking model yielded better results compared to the base classifiers. With the Cleveland data, their stacking model yielded 92%, 92.6%, and 91% for accuracy, precision, and sensitivity, respectively, and 91% for specificity. Albert et al. [11] used a combination of the synthetic minority oversampling technique (SMOTE) and a pruned decision tree to improve classifier performance. Less informative class features were pruned, and hyperparameter settings were optimised.

Rimal and Sharma [12] evaluated various hyperparameter optimisation techniques for random forests and support vector machines using the UCI heart disease dataset. The standard random forest model achieved 86.6% accuracy. With Bayesian optimisation, the random forest model reached 89%, and the support vector machine model reached 90%. Genetic algorithms achieved 86.8–88.5% accuracy, whereas the Optuna framework achieved 84% accuracy when applied to support vector machines. The heart-stroke prediction model combined an artificial neural network with permutation feature importance and local interpretable model-agnostic explanations (LIME), achieving 95% accuracy on clinical stroke data. DEEP CARDIO, an Internet of Things (IoT) system, collects blood pressure, glucose, ECG, and pulse data from wearable sensors developed by Yashudas et al. [13]. A BiGRU model with an attention mechanism was implemented to classify five types of cardiovascular diseases and provide personalised recommendations. The accuracy of the DEEP CARDIO model was 99.90%, which is a significant improvement over MABC SVM (86.91%), HCBDA (88.65%), and MLbPM (93.63%).

Hossain et al. [14] analysed a cross-sectional dataset comprising 391 patients with cardiovascular disease and 260 control patients, collected between August 2022 and April 2023. Logistic regression, Naïve Bayes, decision tree, AdaBoost, random forest, and bagging classifiers were implemented. The random forest classifier achieved 98.04% accuracy, with 96.15% precision, 100% recall, and an area under the curve (AUC) of 0.989, while the logistic regression model achieved the lowest accuracy of 95.42%. El Sofany et al. [15] incorporated the chi-square, ANOVA, and mutual information methods to select pertinent features from public and private datasets. Ten classifiers were examined. Class imbalance was reduced by applying the Synthetic Minority Oversampling Technique (SMOTE). The classification using XGBoost and the selected SF 2 feature subset yielded 97.57% accuracy, 96.61% sensitivity, 90.48% specificity, 95.00% precision, and an AUC of 98%. Explanations based on SHAP were used to interpret the impact of features. Srinivasan et al. used eight classifiers and the UCI dataset to create an active learning technique for predicting CVD [16]. The best-performing classifiers, Naïve Bayes and a radial basis function neural network, achieved 94.78% and 90.78% accuracy, respectively. Learning Vector Quantisation, an active learning technique, achieved 98.7% accuracy, further demonstrating the advantage of active learning in selecting the most informative samples.

Chandrasekhar and Peddakrishna [17] improved heart disease prediction by combining multiple machine learning algorithms and optimisation methods. Metaheuristic search adjusted model hyperparameters, and oversampling methods improved control of class imbalance. Their optimised ensemble model outperformed the rest. Baghdadi et al. [18] proposed novel machine learning methodologies for early diagnosis of cardiovascular disease. By layering optimised methods and deep neural networks on top

of ensemble learning, they identified complex patterns in clinical datasets, achieving higher diagnostic accuracy than standard classifiers. Dalal et al. [19] utilised logistic regression, random forest, support vector machines and gradient boosting to evaluate the risk of developing cardiovascular disease. Their comparative analysis established that in terms of accuracy, precision, recall and F1 score, ensemble frameworks surpassed single classifiers. Khan et al. [20] developed two unique ensembles and blending networks (ENSCVDD Net and BlCVDD Net) for the prediction of heart disease. These networks, engineered by blending within deep learning frameworks to combine predictions from several subordinate models, attained remarkable accuracy compared to others based on classical machine learning techniques. Khan et al. [21] performed a comparative analysis of several classification methods (decision trees, random forests, logistic regression, Naïve Bayes and support vector machines) utilising a clinical dataset from Pakistan. Random forests achieved the highest accuracy of 85.01%, along with the highest sensitivity and area under the curve, ranking it the best among classifiers for predicting cardiovascular disease. Recent studies have also extended machine learning toward connected and explainable healthcare systems. Kumar et al. [22] developed the CDF-DI framework by integrating machine learning with IoT-enabled cardiac monitoring. Kumar et al. [23] proposed LivXAI-Net, combining explainable AI with IoT-supported monitoring for liver disease diagnosis. Mehta and Verma [24] further demonstrated the applicability of machine learning for cancer diagnosis using fine-needle aspirate data. Together, these studies highlight the growing role of predictive, connected, and explainable AI in clinical decision support.

3 Materials and methods

This section presents the materials, preprocessing protocols, mathematical formulation, learned strategy, pseudocode, and evaluation metrics that contributed to the design of the Feature-Weighted Calibrated Tabular Explainable Artificial Intelligence (FWC-TabXAI++) framework. The method aims to yield a calibrated, explainable probability of heart disease diagnosis from heterogeneous clinical tabular data. In doing so, the method preserves interpretability at both the global and individual-patient levels. The method was implemented using the provided experimental code, which included data preprocessing, outlier removal, categorical data encoding, data normalisation, construction of feature interactions, modelling of mutual information, feature selection, heterogeneous model construction, model validation and blending, confidence intervals, and SHAP and LIME analyses.

3.1 Dataset description

The dataset was obtained from the publicly available Kaggle Heart Disease repository and is based on the UCI Heart Disease dataset [25]. The data used for this study are a structured clinical dataset on heart disease, including demographic, electrocardiographic, exercise-related, and biochemical variables. Each record contains a single observation on a patient; the binary response variable indicates whether the patient has heart disease. There are 918 observations and 12 variables in the raw data. A total of 836 observations were used for model development after clinical consistency correction and IQR-based outlier exclusion. The cleaned observations included 454 samples from the heart-disease class and 382 from the non-heart-disease class, indicating that

the classification problem was moderately imbalanced but clinically feasible. The input space for the proposed heart-disease prediction framework is presented in Table 1, and each encoded variable is associated with a clinically interpretable concept.

Table 1 maps each model input to a clinically interpretable physiological, symptomatic, or diagnostic attribute. There are numeric variables (Age, resting blood pressure, cholesterol, maximum heart rate, exercise-induced ST depression) and categorical variables (chest-pain type, resting ECG status, exercise-induced angina, ST-segment slope). A mixed data type is a valid justification for a preprocessing pipeline that includes both numerical standardisation and one-hot encoding of categorical predictors. Let the cleaned clinical dataset be represented as a finite set of patient-level observations. A feature vector and a binary diagnostic label describe each patient. The mathematical representation is given in Eq. (1).

$$\mathrm{D} = \left\{\left(\mathbf{x}_i, \mathrm{y}_i\right)\right\}_{i=1}^{N}, \quad \mathbf{x}_i \in \mathbb{R}^d, \quad y_i \in \{0,1\}, \quad N = 836 \tag{1}$$

In Eq. (1), D denotes the complete cleaned dataset, $\mathbf{x}_i$ is the d-dimensional encoded clinical feature vector of the i-th patient, y_i is the corresponding binary heart-disease label, and N is the number of retained observations after preprocessing. The target value $y_i = 1$ indicates a heart-disease-positive case, whereas $\mathrm{y_i} = 0$ indicates a non-heart-disease case. The class distribution of the cleaned dataset is expressed in Eq. (2). This formulation is useful for defining stratified sampling and for evaluating the diagnostic sensitivity and specificity of the proposed model.

$$N_1 = \sum_{i=1}^{N} \mathbf{1}(y_i = 1) = 454, \quad N_0 = \sum_{i=1}^{N} \mathbf{1}(y_i = 0) = 382 \tag{2}$$

In Eq. (2), $\mathrm{N_1}$ and $\mathrm{N_0}$ denote the number of heart-disease-positive and heart-disease-negative observations, respectively.

Table 1 Clinical attributes used for heart disease prediction

Variable	Clinical attribute	Type	Clinical interpretation
A	Age	Numerical	Age of the patient in years; increasing Age is generally associated with higher cardiovascular risk
S	Sex	Categorical	Biological sex of the patient
C^p	ChestPainType	Categorical	Type of chest pain, including asymptomatic and angina-related categories
B^r	RestingBP	Numerical	Resting blood pressure is clinically related to vascular load and hypertension
C^h	Cholesterol	Numerical	Serum cholesterol concentration is used as a cardiovascular risk marker
F^{bs}	FastingBS	Categorical	Fasting blood sugar status is represented as a binary clinical indicator
E^r	RestingECG	Categorical	Resting electrocardiographic finding
H^m	MaxHR	Numerical	Maximum heart rate achieved during exercise testing
E^a	ExerciseAngina	Categorical	Presence or absence of exercise-induced angina
O^p	Oldpeak	Numerical	Exercise-induced ST depression relative to rest
L^{st}	ST_Slope	Categorical	Slope of the peak exercise ST segment
y	HeartDisease	Binary target	Diagnostic class label, where 1 denotes heart disease, and 0 denotes no heart disease

3.2 Dataset partitioning

Stratified three-way partitioning was used to create the training, validation, and test sets as closely as possible to the original class distribution. The training subset was used to train the model parameters; the validation subset was used to fine-tune the blending weights and decision threshold, whereas the independent test subset was used solely for final performance reporting.

$$D = D^{tr} \cup D^{val} \cup D^{test}, \quad D^{tr} \cap D^{val} \cap D^{test} = \emptyset \tag{3}$$

Equation (3) states that the dataset was separated into mutually exclusive training, validation, and testing partitions. The raw dataset was first stratified into training, validation, and test partitions. After applying the training-fitted preprocessing and outlier criteria unchanged to all partitions, 534 training, 134 validation, and 168 test samples were retained, giving 836 samples in total. A stratified hold-out design was used to maintain a completely independent test subset for final evaluation, while the validation subset was reserved for ensemble-weight and threshold optimisation. To reduce dependence on this single partition, repeated stratified validation was additionally performed and reported in Sect. 4.6.

Table 2 summarises the stratified training, validation, and testing partitions of the cleaned dataset. The training subset was used for model fitting, the validation subset for probability-blending and threshold optimisation, and the independent test subset for final performance reporting.

3.3 Preprocessing protocol

All data-dependent preprocessing steps were fitted using only the training subset. The non-zero replacement means, IQR thresholds, encoding schema, standardization parameters, interaction-feature schema, and mutual-information ranking were then applied unchanged to the validation and independent test subsets. The test set was used only for final performance evaluation. Certain continuous clinical measurements, such as serum cholesterol and resting blood pressure, cannot take a physiologically meaningful value of zero in this diagnostic context. For this reason, each variable containing zeros was treated as an invalid measurement, and the feature's non-zero mean was used instead. The replacement rule is given in Eq. (4).

$$\tilde{x}_{ij} = x_{ij} \quad (x_{ij} \neq 0); \quad \tilde{x}_{ij} = \mu_j^{nz} \quad (x_{ij} = 0) \tag{4}$$

In Eq. (4), x_{ij} is the original value of the j-th continuous clinical variable for the i-th patient,$\tilde{x}_{ij}$ is the corrected value, and μ_j^+ is the mean computed only from non-zero observations of the same variable. This prevents physiologically impossible zero values from biasing the distribution of continuous predictors.

Table 2 Post-preprocessing dataset split summary

Subset	Samples	Class 0	Class 1	Class 0 (%)	Class 1 (%)
Training	534	244	290	45.69	54.31
Validation	134	61	73	45.52	54.48
Testing	168	77	91	45.83	54.17
Total	836	382	454	45.69	54.31

$$\mu_j^{nz} = \frac{1}{n_j^{nz}} \sum_{i \in I_j^{nz}} x_{ij}, \quad I_j^{nz} = \{i : x_{ij} \neq 0\}, \quad n_j^{nz} = \left|I_j^{nz}\right| \tag{5}$$

Equation (5) defines the non-zero mean μ_j^+. The set $\mathcal{J}_j^+$ contains only those samples for which the j-th feature has a valid non-zero value. To reduce the influence of extreme and clinically inconsistent continuous values, outlier filtering was performed using the interquartile range method. For each numerical feature, the first quartile, third quartile, and interquartile range were computed as shown in Eq. (6).

$$Q_{1j} = P_{25}\left(x_j\right), \quad Q_{3j} = P_{75}\left(x_j\right), \quad \mathrm{IQR}_j = Q_{3j} - Q_{1j} \tag{6}$$

In Eq. (6), P_{25} and P_{75} denote the 25th and 75th percentiles, respectively. The lower and upper admissible bounds for each continuous feature are then defined in Eq. (7).

$$L_j = Q_{1j} - 1.5\,\mathrm{IQR}_j, \quad U_j = Q_{3j} + 1.5\,\mathrm{IQR}_j \tag{7}$$

A sample was retained only if all selected numerical features were within their corresponding IQR limits. This decision rule is mathematically expressed in Eq. (8).

$$\mathrm{D}^c = \left\{(\mathbf{x}_i, y_i) \in \mathrm{D} : L_j \leq x_{ij} \leq U_j,\ \forall j \in J_n\right\} \tag{8}$$

In Eq. (8), D^c denotes the cleaned dataset after outlier filtering, and J_n is the set of numerical clinical variables. In the implemented experiment, the IQR procedure removed 27 outliers from RestingBP, 40 from Cholesterol, 2 from MaxHR, and 13 from Oldpeak, while Age did not produce outlier removal. After the training-derived preprocessing criteria were applied consistently across the three partitions, 836 of the original 918 samples were retained. Categorical variables were transformed into binary indicator vectors using one-hot encoding. For a categorical feature z_j with K_j possible categories, the one-hot mapping is given in Eq. (9).

$$\phi_j\left(z_{ij}\right) = \left[\mathbf{1}\left(z_{ij} = c_{j1}\right),\ \mathbf{1}\left(z_{ij} = c_{j2}\right), \ldots, \mathbf{1}\left(z_{ij} = c_{jK_j}\right)\right] \tag{9}$$

Equation (9) converts a non-ordinal categorical feature into a binary vector without imposing artificial numerical ordering among categories. This is important for variables such as chest pain type, resting ECG category, exercise-induced angina, and ST-segment slope. Each numerical variable was standardised using the training-set mean and standard deviation. The standardised value is defined in Eq. (10).

$$x_{ij}^z = \frac{x_{ij} - \mu_j^{\mathrm{tr}}}{\sigma_j^{\mathrm{tr}}} \tag{10}$$

In Eq. (10), x_{ij}^z is the standardised value, μ_j^{tr} is the mean of the j-th feature estimated from the training partition, and σ_j^{tr} is its training-set standard deviation. Estimating scaling parameters only from the training subset avoids information leakage from the validation and test data.

3.4 Proposed FWC-TabXAI++ method

Figure 1 presents the complete architecture of the proposed FWC-TabXAI++ framework. The model is designed as a staged clinical decision-support pipeline rather than as a single black-box classifier. The input consists of structured clinical variables, which

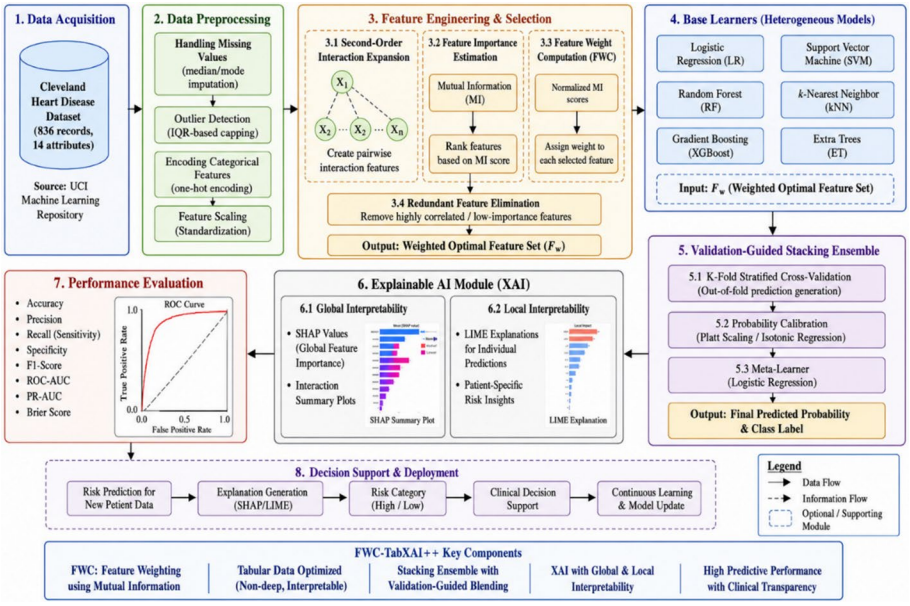


Fig. 1 Overall architecture of the proposed FWC-TabXAI++ framework

are first cleaned, encoded, and standardised, then transformed into an enhanced representation using feature-interaction expansion and mutual-information-based selection. The central methodological contribution is the heterogeneous learner pool followed by validation-based probability blending. Instead of selecting only one classifier, the framework trains multiple complementary learners, including linear, kernel-based, tree-based, boosting-based, and neural components. Their probability outputs are combined using optimised validation-derived weights, allowing the final model to exploit different decision boundaries and data representations. The calibration block in Fig. 1 represents an optional post-hoc stage. In this study, calibration was assessed using standard calibration metrics and reliability analysis. The final prediction layer applies threshold optimisation and is followed by statistical validation and explainability modules. SHAP and LIME explanations are placed after the predictive layer because they translate the model output into clinically interpretable feature contributions.

Figure 2 illustrates the experimental workflow used to evaluate the proposed model. The workflow starts with raw data and stratified partitioning, followed by training-fitted preprocessing, feature construction, model development, validation tuning, and independent test evaluation. This makes the pipeline reproducible and avoids ambiguity about whether preprocessing was performed before or after data splitting. The modelling stage compares 11 conventional machine-learning classifiers with the proposed FWC-TabXAI++ framework. The validation set is used to determine ensemble blending weights and the operating threshold, while the independent test set is reserved for final reporting. This separation is methodologically important because tuning on the test set would artificially inflate model performance. Interaction among clinical variables may be nonlinear. The sample counts shown after partitioning correspond to the retained post-preprocessing samples rather than the initial raw split sizes. The significance of max HR can vary by Age, and ST-segment depression can be synergistic with EIA. To explicitly show such interactions, the proposed framework uses a 2nd-order interaction mapping. The new feature vector is denoted as in Eq. (11).

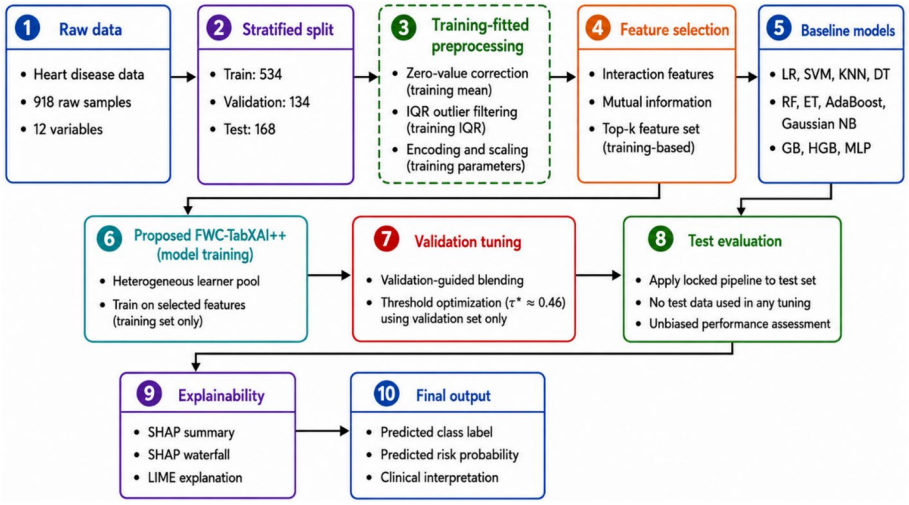


Fig. 2 Simplified experimental workflow of the proposed study

$$\Phi(\mathbf{x}_i) = \left[x_{i1}, \ldots, x_{id},\ x_{i1}x_{i2},\ x_{i1}x_{i3}, \ldots, x_{i(d-1)}x_{id}\right] \tag{11}$$

Equation (11) augments the original clinical feature vector with pairwise interaction terms. The mapping enables conventional learners to access multiplicative relationships among clinical variables without requiring a fully deep architecture. If the original encoded dimension is d, the maximum number of interaction-expanded variables is given in Eq. (12).

$$d_\Phi = d + \frac{d(d-1)}{2} \tag{12}$$

Equation (12) shows that interaction expansion increases the dimensionality of the tabular representation. Therefore, a feature selection step is required to control redundancy and prevent overfitting. The proposed method uses mutual information between target and expanded features to assign relevance to the expanded features. Mutual information quantifies how much uncertainty about the class label is eliminated with the observation of a feature. The MI score between an expanded feature V_j and class label y is defined in Eq. (13).

$$I(v_j; y) = \sum_{v \in V_j} \sum_{y \in \{0,1\}} p(v, y) \log\left(\frac{p(v, y)}{p(v)\, p(y)}\right) \tag{13}$$

In Eq. (13), $I(v_j; y)$ quantifies the dependence between the j-th expanded feature and the target label. A higher value indicates stronger diagnostic relevance. The selected feature subset is defined in Eq. (14).

$$S_k = \arg\operatorname*{top}_k \left\{I(v_j; y) : j = 1, 2, \ldots, d_\Phi\right\} \tag{14}$$

The most informative expanded clinical variables are retained in Eq. (14). The implementation limited the number of selected variables to 60, and k was restricted to the available expanded feature dimension. This step is part of the feature-weighted component of FWC-TabXAI++, where candidate learners are given a compact representation of features that highlight those that are diagnostically informative, including feature

interactions. The selected feature representation is provided to a heterogeneous pool of candidate learners. The pool includes linear, kernel-based, tree-based, boosting-based, and neural-network-based classifiers. Let M denote the number of candidate learners and let $f_m(\cdot)$ be the m-th learner. Each learner estimates a posterior probability for the positive heart-disease class as shown in Eq. (15).

$$p_m(\mathbf{x}_i) = f_m\left(\Phi_{S_k}(\mathbf{x}_i)\right) \approx P\left(y_i = 1|\mathbf{x}_i\right), \quad m = 1, 2, \ldots, M \tag{15}$$

In Eq. (15), $\Phi_{S_k}(\mathbf{x}_i)$ is the selected interaction-aware feature vector for the i-th sample. Using heterogeneous learners improves robustness because different model families capture different decision structures: linear separability, nonlinear margin behavior, hierarchical feature partitioning, boosting-based residual correction, and neural approximation. The proposed framework does not use one of the best classifiers; instead, it learns a convex combination of candidate probability outputs from the validation subset. The blended diagnostic probability is defined as in Eq. (16).

$$\hat{p}_i = \sum_{m=1}^{M} w_m p_m(\mathbf{x}_i), \quad w_m \geq 0, \quad \sum_{m=1}^{M} w_m = 1 \tag{16}$$

In Eq. (16), w_m is the contribution weight of the m-th candidate learner. The non-negativity and unit-sum constraints ensure that the final score remains a valid probability. The weight vector lies on the probability simplex Δ^M, defined in Eq. (17).

$$\mathbf{w} \in \Delta^M = \left\{ \mathbf{w} \in \mathbb{R}^M : w_m \geq 0, \sum_{m=1}^{M} w_m = 1 \right\} \tag{17}$$

The optimal blending weights were identified using maximum validation performance. In implementation, the random simplex search was done with Dirichlet-distributed candidate weight vectors. The objective of the empirical validation is written as Eq. (18).

$$\mathbf{w}^\star = \arg\max_{\mathbf{w} \in \Delta^M} J^{\text{val}}(\mathbf{w}, \tau) \tag{18}$$

In Eq. (18), $J^{\text{val}}(\mathbf{w}, \tau)$ denotes the validation objective evaluated after applying threshold τ to the blended probabilities. The objective prioritises accuracy and resolves ties using the F1-score. This design improves diagnostic decision quality while retaining a probabilistic output for calibration and uncertainty analysis. The default threshold of 0.50 might not be the best threshold if the costs of false positives and false negatives are not equal. Hence, FWC-TabXAI++ fine-tunes the decision threshold on the validation data. The binary decision rule is given in Eq. (19).

$$\hat{y}_i(\tau) = \begin{cases} 1, & \text{if } \hat{p}_i \geq \tau, \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

In Eq. (19), τ is the diagnostic threshold. The threshold optimisation problem is given in Eq. (20).

$$\tau^\star = \arg\max_{\tau \in T} J^{\text{val}}(\mathbf{w}^\star, \tau) \tag{20}$$

From Eq. (20), the value of the threshold that maximises the validation criterion is chosen. The threshold used in the experiment is around 0.46. This value was then used to calculate the final diagnostic labels based on the independent test probabilities. After validation-based optimisation, the candidate learners are refitted using the combined training and validation subsets, and the learned blending weights are applied to the independent test subset. The final probability estimator is expressed in Eq. (21).

$$F(\mathbf{x}) = \sum_{m=1}^{M} w_m^{\star} f_m\left(\Phi_{S_k}(\mathbf{x})\right) \tag{21}$$

Equation (21) defines the final FWC-TabXAI++ model. The predicted diagnostic label is obtained by applying the optimised threshold to $F(x)$, as shown in Eq. (22).

$$\hat{y} = 1\left(F(x) \geq \tau^{\star}\right) \tag{22}$$

3.5 Justification of the proposed formulation

Proposition 1 A linear classifier trained on the interaction-expanded representation Φ() can represent any second-order decision function of the original encoded variables.

Proof Consider a second-order diagnostic scoring function of the form shown in Eq. (23).

$$g(\mathbf{x}) = \beta_0 + \sum_{j=1}^{d} \beta_j x_j + \sum_{j<l} \beta_{jl} x_j x_l \tag{23}$$

The expanded representation contains all original variables x_j and all pairwise products $x_j x\ell$ for $j < \ell$. Therefore, Eq. (23) can be rewritten as an inner product in the expanded space, as shown in Eq. (24).

$$g(\mathbf{x}) = \beta_0 + \beta_{\Phi}^{T} \Phi(\mathbf{x}) \tag{24}$$

After interaction expansion, Eq. (24) can be proven to represent the second-order relationships of the original clinical feature space linearly. Thus, simpler learners can be used to model clinically relevant feature interactions without the need for deep nonlinear architectures when using the proposed representation. This defines a mathematical motivation for the feature-interaction stage.

Proposition 2 Under the same validation objective J^{val}, the optimised convex ensemble is at least as good as the best candidate learner on the validation subset, provided that the simplex search includes the simplex vertices or can approximate them sufficiently.

Proof The m-th individual learner can be represented as a vertex vector $\mathbf{e}_m$ of the simplex, where $e_m = 1$ for the selected learner and zero for all other learners. Since every $\mathbf{e}_m$ belongs to Δ^M, the feasible set of the ensemble optimisation contains all individual learners. Therefore, Eq. (25) holds.

$$\max_{\mathbf{w} \in \Delta^M} J^{val}(\mathbf{w}, \tau) \geq \max_{m} J^{val}(\mathbf{e}_m, \tau) \tag{25}$$

If the optimiser traverses the simplex region relevant to the validation criterion in Eq. (25), then the validation-optimised ensemble cannot be inferior to the best single member, measured by the same validation criterion, as shown in the following equation. This is a reason to use a probability blending approach over choosing one classifier. The statement is an empirical optimisation property, which in itself does not provide evidence for external clinical generalisation, but rather is assessed by independent testing and confidence-interval analysis.

Proposition 3 If each candidate model returns a valid probability $p_m(\mathbf{x}) \in [0, 1]$, then the blended output $\hat{p}$ is also a valid probability.

Proof Since $0 \le p_m(\mathbf{x}) \le 1$ and $w_m \ge 0$ for all m, the lower bound follows directly from Eq. (26).

$$\hat{p} = \sum_{m=1}^{M} w_m p_m(\mathbf{x}) \ge 0 \tag{26}$$

The upper bound follows because each $p_m(\mathbf{x})$ is at most one, as shown in Eq. (27).

$$\hat{p} = \sum_{m=1}^{M} w_m p_m(\mathbf{x}) \le \sum_{m=1}^{M} w_m = 1 \tag{27}$$

From Eqs. (26) and (27), $0 \le \hat{p} \le 1$. Thus, the convex blend preserves probabilistic validity. This property is essential because the final output is used for calibrated risk estimation, thresholding, uncertainty grouping, and explainability. Calibration is evaluated using the Brier score, which measures the mean squared discrepancy between predicted probabilities and true binary labels. The Brier score is given in Eq. (28).

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} (\hat{p}_i - y_i)^2 \tag{28}$$

The lower the Brier score, the better the probability calibration. FWC-TabXAI++ yields a convex probability estimate so that the final probabilities can be directly used in the computation of the Brier score. The framework is not limited to hard classification but also offers a risk score which can be used for clinically interpretable decision support.

3.6 Explainability formulation

SHAP was used to quantify feature-level contribution to model predictions. For a prediction function F and feature set, the Shapley value of feature j is defined in Eq. (29).

$$\phi_j = \sum_{A \subseteq P,\, j \notin A} \frac{a!\,(p-a-1)!}{p!} \left[F\left(A \cup \{j\}\right) - F(A) \right], \quad a = |A|,\ p = |P| \tag{29}$$

In Eq. (29), ϕ_j denotes the contribution of the j-th feature, A is a subset of features not containing j, and $F\left(A \cup \{j\}\right) - F(A)$ measures the marginal effect of adding feature j to subset A. The SHAP decomposition for an individual prediction is given in Eq. (30).

$$F(\mathbf{x}_i) = E[F(\mathbf{X})] + \sum_{j=1}^{d} \phi_{ij} \tag{30}$$

The prediction for an individual patient can be broken down into the average model prediction and additive contributions from the features in Eq. (30). The positive SHAP values will add to the predicted risk of heart disease, and the negative SHAP values will subtract from it. LIME was used to provide a locally faithful approximation of the model around a representative test instance. For a patient instance $\mathbf{x}_i$, LIME constructs an interpretable surrogate model h by minimising the locally weighted loss shown in Eq. (31).

$$\mathrm{h}^{\star} = \arg\min_{\mathrm{h}\in\mathcal{H}} \mathcal{L}(F, \mathrm{h}, \pi_{\mathbf{x}_i}) + \Omega(\mathrm{h}) \tag{31}$$

In Eq. (31), $\mathrm{h}^{\star}$ measures the discrepancy between the original model F and the surrogate explanation model h, and Ω(h) penalises excessive explanation complexity. Thus, LIME complements SHAP by providing a sparse, patient-specific explanation of the diagnostic decision.

3.7 Algorithms

The method proposed is summarised in two pseudocode blocks. The training procedure is listed in Algorithm 1, which involves preprocessing, candidate model fitting, validation-based weight optimisation, and final refitting. Inference and explainability for a new/hold-out patient sample are given in Algorithm 2.

Algorithm 1 Training procedure for FWC-TabXAI++

Component	Description
Input	Raw dataset, numerical feature set, categorical feature set, candidate learner pool $\{f_1,\ldots,f^M\}$, threshold grid, number of selected features k
Output	Trained FWC-TabXAI++ model F(·), optimal blending weights, optimal threshold τ*
1	Split the raw dataset into training, validation, and test subsets using stratified sampling
2	Fit zero-value correction and IQR thresholds using the training subset only in Eqs. (6) to (8)
3	Apply the fitted preprocessing pipeline unchanged to validation and test subsets
4	Apply one-hot encoding to categorical variables according to Eq. (9)
5	Standardise numerical variables using training-set statistics according to Eq. (10)
6	Generate second-order interaction features as defined in Eq. (11)
7	Compute mutual-information scores using Eq. (13) and retain the top-k variables using Eq. (14)
8	Train each candidate learner f_m on the selected training representation
9	Generate validation probabilities for all candidate learners using Eq. (15)
10	Search the probability simplex Δ^M and threshold grid using Eqs. (18) and (20)
11	Refit all candidate learners on the combined training and validation subsets
12	Construct the final probability estimator according to Eq. (21)

Algorithm 2 Inference and explainability procedure for FWC-TabXAI++

Component	Description
Input	Trained model F(·), optimised threshold τ*, fitted preprocessing operators, patient feature vector
Output	Predicted probability, diagnostic label $\hat{y}^*$, SHAP explanation, and LIME explanation
1	Apply the fitted zero-value correction, categorical encoding, and numerical standardisation operators
2	Generate the selected interaction-aware representation
3	Compute candidate learner probabilities for m = 1, 2, …, M
4	Compute the blended diagnostic probability

Algorithm 2 Inference and explainability procedure for FWC-TabXAI++

Component	Description
5	Assign the predicted label $\hat{y}^*$
6	Estimate SHAP contributions φ_j to explain global and patient-specific feature influence
7	Fit a local LIME surrogate h* around h* using Eq. (31)
8	Report probability, class label, dominant positive contributors, dominant negative contributors, and uncertainty status

3.8 Evaluation metrics

Confusion-matrix-based diagnostic metrics, discrimination metrics and probability-calibration metrics were all used to test the independent test set. Suppose that TP, TN, FP and FN represent the numbers of true positives, true negatives, false positives, and false negatives, respectively.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{32}$$

Accuracy in Eq. (32) is the percentage of the number of patients correctly classified of the total number of test patients. While it is a valuable indicator of total behaviour, it fails to distinguish false-negative from false-positive behaviour, and so is not adequate as a stand-alone diagnostic tool.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{33}$$

The precision of Eq. (33) describes the percentage of patients that the model predicts as heart disease positive who actually are heart disease positive. The aim of high precision is to minimise unwanted clinical concerns as a result of false-positive predictions.

$$\text{Recall} = \text{Sensitivity} = \frac{TP}{TP + FN} \tag{34}$$

Recall, also called sensitivity, is given in Eq. (34). It measures the ability of the model to correctly detect heart-disease-positive cases. In clinical screening, high sensitivity is important because false negatives may delay diagnosis and treatment.

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{35}$$

Specificity in Eq. (35) measures the ability of the model to correctly identify non-heart-disease cases. High specificity reduces the number of healthy patients incorrectly flagged as diseased.

$$\text{F}_1\text{-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{36}$$

The F1 score in Eq. (36) is the harmonic mean of precision and recall. It is useful when both false-positive and false-negative errors must be considered simultaneously.

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \tag{37}$$

The true-positive rate and false-positive rate in Eq. (37) are the metrics to plot a receiver operating characteristic (ROC) curve. This curve area is called ROC-AUC, measuring discrimination at all possible thresholds.

$$ROC - AUC = \int_0^1 TPR\,(FPR)\; d\,(FPR) \tag{38}$$

ROC-AUC in Eq. (38) summarises the ability of the model to rank positive cases higher than negative cases. A value close to one indicates strong class separability.

$$\text{PR - AUC} = \int_0^1 \text{Precision}\,(\text{Recall})\; d\,(\text{Recall}) \tag{39}$$

The precision-recall trade-off over thresholds is summarised in PR-AUC in Eq. (39). It can be useful if the class distribution is unbalanced and/or the positive clinical condition is the focus.

$$\text{BS} = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - y_i)^2 \tag{40}$$

The Brier score in Eq. (40) evaluates probabilistic calibration by measuring the squared error between predicted probabilities and observed outcomes. Lower values indicate better-calibrated risk estimates.

$$\text{CI}_{95\%}\,(\theta) = \left[P_{2.5}\left(\theta^b\right),\; P_{97.5}\left(\theta^b\right)\right] \tag{41}$$

The bootstrap confidence interval for a performance statistic θ is given by Eq. (41). Empirical distribution of accuracy, F1 score and ROC AUC were estimated via repeated resampling from the data and 95% confidence intervals were reported as the 2.5th and 97.5th percentiles.

4 Results and discussion

This section presents experimental results for the proposed FWC-TabXAI++ framework and benchmarking machine-learning models for comparison. The outcomes of preprocessing and exploratory statistics are analysed through classification, ablation, uncertainty, and interpretability tests. The results are framed in a context of diagnostic trust, the calibration of predictions, and a clinical rationale for interpretability.

4.1 Data preprocessing and exploratory analysis

Table 3 presents the results of the outlier detection strategy using the IQR for the continuous clinical variables. The interquartile range method defines the lower and upper acceptance limits as Q1 − 1.5 × IQR and Q3 + 1.5 × IQR, respectively. Samples with values outside of these ranges were subsequently discarded in order to minimise the effects of extreme values on the model fit. No outlier removal was seen with Age, indicating

Table 3 IQR-based outlier detection and cleaning summary

Feature	Q1	Q3	IQR	Lower bound	Upper bound	Removed outliers
Age	47.00	60.00	13.00	27.50	79.50	0
RestingBP	120.00	140.00	20.00	90.00	170.00	27
Cholesterol	214.00	265.50	51.50	136.75	342.75	40
MaxHR	119.50	156.00	36.50	64.75	210.75	2
Oldpeak	0.00	1.50	1.50	−2.25	3.75	13
Dataset size	Before: 918	After: 836				82 total removed

Table 4 Chi-square association analysis of categorical clinical variables

Feature	Chi-square value	*p*-value	Degrees of freedom	Interpretation
Sex	82.5704	1.0197e−19	1	Significant
ChestPainType	251.4363	3.2002e−54	3	Significant
FastingBS	65.3338	6.3227e−16	1	Significant
RestingECG	11.1938	0.0037	2	Significant
ExerciseAngina	203.7128	3.2333e−46	1	Significant
ST_Slope	317.5583	1.1043e−69	2	Significant

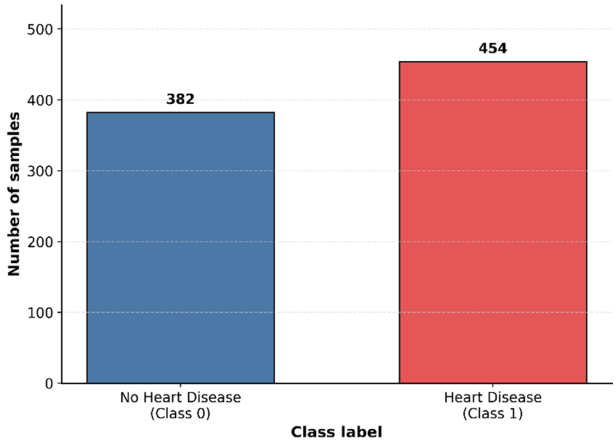


Fig. 3 Class distribution of the heart disease dataset

that its distribution was within the range expected for the cohort. Resting blood pressure, cholesterol, maximum heart rate, and oldpeak, on the other hand, had values out of the IQR ranges. After numerical outlier cleaning, the size of the dataset was reduced from 918 to 836 samples, and 82 records were omitted.

The results of the chi-square test of independence between each categorical clinical variable and the target class are shown in Table 4. The results of this analysis support the statistical evidence that the categorical predictors are not randomly distributed with respect to heart-disease status. All the categorical attributes evaluated are statistically significant at $p < 0.05$. ST_Slope, ChestPainType, and ExerciseAngina are categorical variables with very strong associations with the target. This is clinically plausible because the ST-segment changes are directly associated with cardiovascular stress response, the angina status and the characteristics of chest pain are directly associated with ischemic risk. Other significant relationships are seen with sex and fasting blood sugar, and ECG at rest provides further diagnostic information. These results connect exploratory clinical analysis with predictive modelling and support the inclusion of categorical predictors in the proposed framework. It makes sense to keep these categorical features in the model, and the following explanation using SHAP/LIME shows that local and global features such as ST-slope and exercise-angina are influential.

Figure 3 shows the post-cleaning distribution of the two outcome classes. The retained dataset contains 382 class-0 records and 454 class-1 records, corresponding to a moderately balanced binary classification problem. This balance reduces the likelihood that high accuracy is caused only by majority-class prediction. Although the imbalance is not severe, the class distribution still motivates the use of sensitivity, specificity, F1-score, PR-AUC, and confusion-matrix-based reporting.

Figure 4 visualises the distribution of the main numerical clinical variables. These variables are important because they represent continuous physiological measurements that may contain skewness, outliers, or clinically extreme observations. The distributional inspection supports the use of robust preprocessing before model training. Resting blood pressure, cholesterol, maximum heart rate, and oldpeak required outlier filtering, while Age was retained without removal. This indicates that preprocessing was feature-specific rather than arbitrary. Such feature-level inspection is important in medical machine learning because preprocessing decisions can influence both model performance and clinical interpretability.

Figure 5 presents the correlation heatmap among numerical clinical variables and the heart-disease target. Correlation analysis is used as an exploratory statistical tool to identify linear relationships, possible redundancy, and clinically plausible feature associations. It does not claim causality but provides useful preliminary insight before nonlinear modelling. The heatmap helps determine whether any variables are strongly collinear and whether individual numerical variables show directional association with the target. Since heart disease is influenced by nonlinear and interaction effects, the proposed framework does not rely solely on correlations; instead, it uses polynomial interaction expansions and nonlinear learners to capture more complex feature relationships.

4.2 Experimental setup

The main hyperparameter values for the baseline models and the proposed framework are summarised in Table 5. This table is crucial for the reproducibility of the results because the baseline comparison would otherwise look under-specified or selectively optimised. The baseline pool includes linear, kernel-based, instance-based, probabilistic, tree-based, boosting-based, and neural network classifiers. This wide range of model selection allows comparison of the proposed method with a variety of learning paradigms, not just a small set of weak ones. Decision Tree and Random Forest were included as representative tree-based methods because such models are widely used for structured clinical data and provide a relevant comparison with the proposed ensemble framework. Several classifiers use class-weight balancing in order to minimise class-distribution effects. The main methodological elements of the FWC-TabXAI++ are recorded in the table: polynomial interaction expansion, mutual-information-based

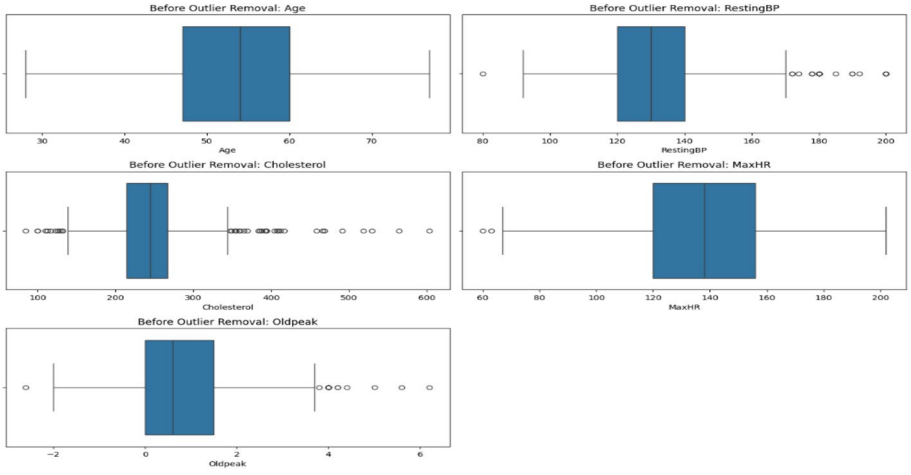


Fig. 4 Distribution of important clinical variables

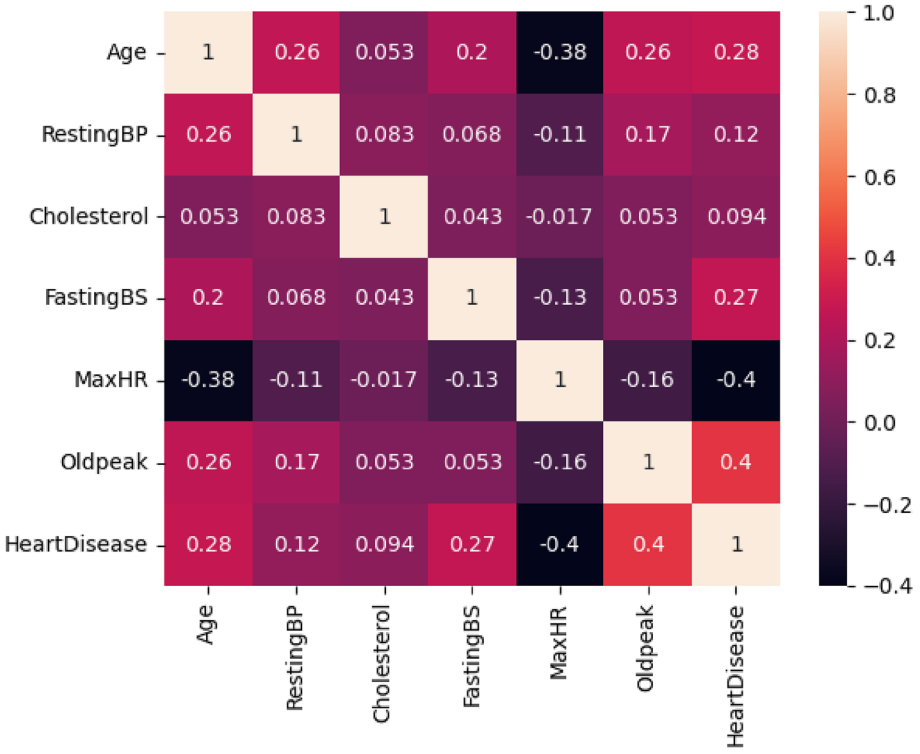


Fig. 5 Correlation heatmap of numerical clinical variables

Table 5 Hyperparameter settings of baseline and proposed models

Model	Main hyperparameter settings
Logistic regression	max_iter = 2000, class_weight = balanced, random_state = 42
SVM	RBF kernel, probability = True, class_weight = balanced, random_state = 42
KNN	n_neighbors = 7
Gaussian Naïve Bayes	Default GaussianNB configuration
Decision tree	max_depth = 5, class_weight = balanced, random_state = 42
Random forest	n_estimators = 300, class_weight = balanced, random_state = 42
Extra trees	n_estimators = 300, class_weight = balanced, random_state = 42
AdaBoost	n_estimators = 200, learning_rate = 0.05, random_state = 42
Gradient boosting	n_estimators = 200, learning_rate = 0.05, max_depth = 3, random_state = 42
Histogram gradient boosting	max_iter = 200, learning_rate = 0.05, max_leaf_nodes = 31, random_state = 42
MLP	hidden layers = (128, 64, 32), max_iter = 800, early_stopping = True
FWC-TabXAI++	Polynomial interaction features, SelectKBest mutual information, seven learner candidates, validation-based probability blending, threshold optimisation

feature selection, heterogeneous candidate learners, validation-based probability weighting, and threshold optimisation.

4.3 Comparative performance analysis

Table 6 shows the main comparative results. It evaluates the proposed FWC-TabXAI++ model against 11 baseline classifiers using discrimination, diagnostic, and probability-quality metrics. The proposed model achieved 91.07% accuracy, 92.22% precision, 91.21% recall, 90.91% specificity, 91.71% F1-score, 94.01% ROC-AUC, 93.24% PR-AUC, and a Brier score of 9.08%. The result shows that the proposed model achieves balanced improvement in detecting positive instances and excluding negative ones. A high recall

Table 6 Performance comparison of proposed FWC-TabXAI++ with baseline models

Model	Acc (%)	Prec (%)	Rec (%)	Spec (%)	F1 (%)	ROC-AUC (%)	PR-AUC (%)	Brier (%)	TP	TN	FP	FN
Proposed FWC-TabXAI++	91.07	92.22	91.21	90.91	91.71	94.01	93.24	9.08	83	70	7	8
SVM	89.29	91.01	89.01	89.61	90.00	93.34	92.81	9.46	81	69	8	10
Gradient boosting	88.69	90.91	87.91	89.61	89.39	94.09	94.19	9.22	80	69	8	11
Gaussian Naïve Bayes	87.50	91.67	84.62	90.91	88.00	94.45	94.96	11.35	77	70	7	14
Random forest	86.90	90.59	84.62	89.61	87.50	93.14	92.83	10.43	77	69	8	14
AdaBoost	86.90	88.76	86.81	87.01	87.78	92.78	93.43	12.78	79	67	10	12
Simple tabular neural network	86.90	87.91	87.91	85.71	87.91	91.88	92.00	10.74	80	66	11	11
Histogram gradient boosting	86.31	88.64	85.71	87.01	87.15	93.61	92.42	9.43	78	67	10	13
Logistic regression	84.52	89.16	81.32	88.31	85.06	93.36	93.80	10.86	74	68	9	17
Extra trees	83.93	87.21	82.42	85.71	84.75	93.21	93.21	10.49	75	66	11	16
KNN	83.33	85.39	83.52	83.12	84.44	91.73	89.98	11.07	76	64	13	15
Decision tree	79.17	87.84	71.43	88.31	78.79	85.19	82.61	15.91	65	68	9	26

rate of 91.21% indicates that the model detected most heart disease cases, and a high specificity rate of 90.91% shows that the model also had a strong ability to identify non-disease cases. This balance is crucial as a diagnostic screening model must not achieve a high degree of sensitivity at the expense of making too many false-positive decisions. The table also shows that Gradient Boosting and Gaussian Naïve Bayes achieved competitive AUC values; however, the proposed model provides the best overall diagnostic operating point after probability blending and threshold optimisation.

Figure 6 shows the ROC curves for all baseline models and the proposed FWC-TabXAI++ framework. ROC analysis evaluates sensitivity against the false-positive rate across all possible thresholds. The proposed model achieved an ROC-AUC of 94.01%, indicating strong discrimination between heart-disease and non-heart-disease cases. The ROC curve is particularly important because it separates threshold-independent ranking quality from a single threshold-based decision. A model with strong ROC-AUC can rank high-risk patients above low-risk patients even when the final operating threshold is adjusted for a specific clinical setting. This supports the use of validation-based threshold optimisation in the proposed framework. Although Gaussian Naïve Bayes achieved a slightly higher ROC-AUC (94.45%) than FWC-TabXAI++ (94.01%), the proposed model obtained the highest accuracy (91.07%) and F1-score (91.71%). Therefore, its advantage is interpreted as balanced overall performance rather than superiority on every individual metric. The small differences in ROC-AUC among FWC-TabXAI++, Gaussian Naïve Bayes, and Gradient Boosting were not interpreted as evidence of statistical superiority.

Figure 7 presents the precision-recall curves for all models. Precision-recall analysis focuses on positive-class prediction and is therefore highly relevant to heart disease detection, where the positive class corresponds to the presence of disease. The proposed model achieved a PR-AUC of 93.24%, showing stable positive-class performance across

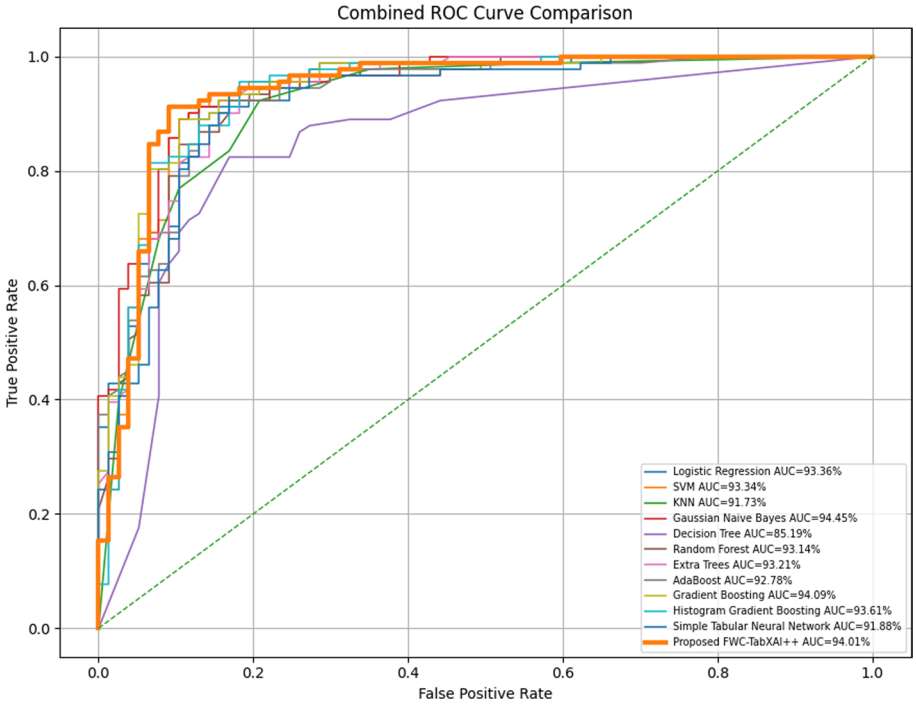


Fig. 6 ROC curve comparison of all models

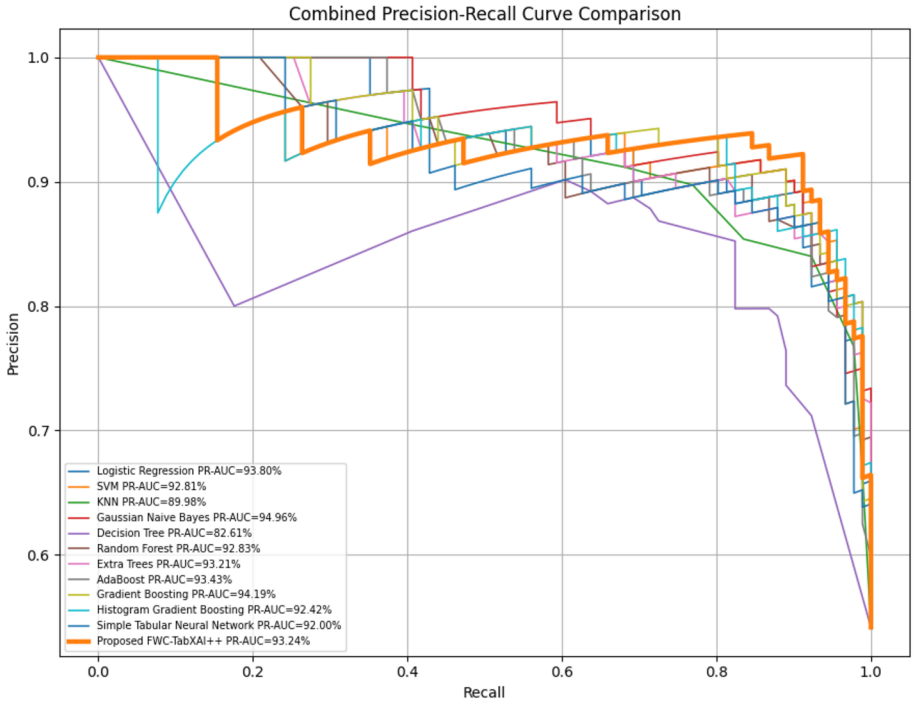


Fig. 7 Precision-recall curve comparison of all models

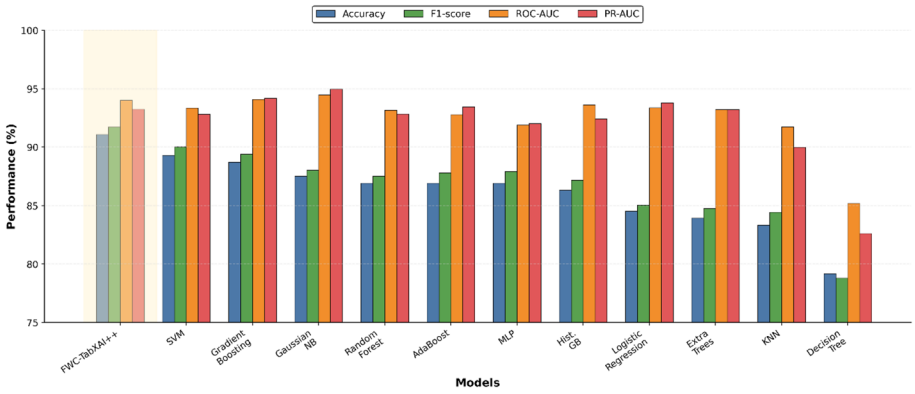


Fig. 8 Overall performance comparison of all models

recall levels. The PR curve complements the ROC curve because precision directly reflects the proportion of predicted disease cases that are true disease cases. In a clinical screening workflow, low precision may lead to unnecessary referrals, repeat tests, and patient anxiety, while low recall may result in missed at-risk patients. The PR curve therefore provides a more clinically interpretable view of positive-case reliability.

Figure 8 compares the main performance metrics across all models, including accuracy, F1-score, ROC-AUC, and PR-AUC. This grouped visualisation provides an integrated view of classification accuracy, positive-class balance, ranking discrimination, and precision-recall behaviour. The proposed FWC-TabXAI++ model shows the strongest overall balance, achieving 91.07% accuracy and 91.71% F1-score while maintaining high ROC-AUC and PR-AUC. Some baseline models perform competitively on individual

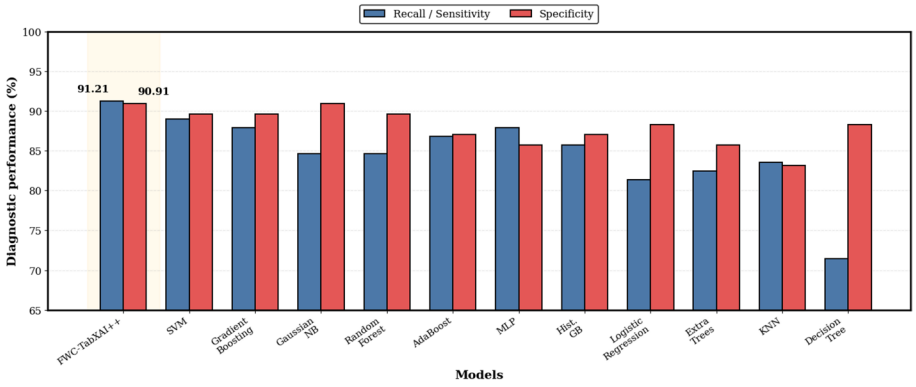


Fig. 9 Recall and specificity comparison

Table 7 Component-wise ablation study of FWC-TabXAI++

Variant	Interaction features	MI selection	Ensemble blending	Threshold optimization	Acc (%)	F1 (%)	ROC-AUC (%)
Best single baseline	No	No	No	No	88.69	89.47	92.86
+ Interaction features	Yes	No	No	No	89.29	90.01	93.21
+ MI feature selection	Yes	Yes	No	No	89.88	90.54	93.55
+ Ensemble blending	Yes	Yes	Yes	No	89.29	89.89	94.01
Full FWC-TabXAI++	Yes	Yes	Yes	Yes	91.07	91.71	94.01

metrics, but the proposed method provides a more consistent profile across threshold-dependent and threshold-independent measures.

4.4 Diagnostic operating-point analysis

This subsection studies the diagnostic balance of the proposed model at the chosen operating threshold. Recall and specificity are examined in tandem to determine the efficacy of the model's detection of heart disease positive cases and the model's ability to exclude non-heart disease-positive cases. This analysis is imperative for evaluating the proposed model's clinical applicability beyond its overall accuracy.

Figure 9 compares recall and specificity for the evaluated models. This is one of the most clinically important visualisations because recall represents the model's ability to detect heart-disease cases, whereas specificity represents its ability to exclude non-disease cases correctly. The proposed FWC-TabXAI++ model achieved 91.21% recall and 90.91% specificity, indicating a balanced diagnostic operating point. This balance is important because excessive emphasis on recall can increase false positives, while excessive emphasis on specificity can miss patients with disease. The proposed threshold provides a clinically reasonable compromise between these outcomes.

4.5 Ablation study

Table 7 presents the component-wise ablation analysis of FWC-TabXAI++. Interaction modelling and MI-based feature selection progressively refine the predictive representation, while ensemble blending improves overall discrimination. Threshold optimization further improves the classification operating point, increasing accuracy and F1-score without altering the ranking-based ROC-AUC and PR-AUC. Overall, the full framework

Table 8 Repeated stratified validation results of FWC-TabXAI++

Metric	Mean (%)	SD (%)	95% CI (%)
Accuracy	85.85	2.50	82.08–90.71
F1-score	87.33	2.14	84.31–91.20
ROC-AUC	92.93	1.64	89.94–95.58
PR-AUC	92.80	1.89	89.43–95.63

Table 9 Bootstrap confidence interval of the proposed model

Metric	Mean (%)	Lower 95% CI (%)	Upper 95% CI (%)
Accuracy	91.04	86.31	95.24
F1-score	91.65	87.01	95.48
ROC-AUC	93.98	89.64	97.51

achieves the best balanced performance, indicating the complementary contribution of its individual components.

4.6 Repeated stratified validation analysis

To reduce reliance on a single data split, repeated stratified validation was performed for the proposed FWC-TabXAI++ framework. In each run, class proportions were preserved, and all preprocessing, feature selection, blending, and threshold selection steps were fitted within the training/validation portion only. The repeated results provide a more reliable estimate of model stability across different patient partitions.

The repeated stratified validation results in Table 8 confirm that the proposed framework remains stable across different patient partitions. The model achieved mean values of 85.85% accuracy, 87.33% F1-score, 92.93% ROC-AUC, and 92.80% PR-AUC. The relatively small standard deviations indicate consistent discrimination and positive-class prediction across repeated runs. The mean selected threshold was 0.462, which is close to the threshold used in the independent test evaluation.

4.7 Statistical validation and uncertainty analysis

The variant without threshold optimisation achieved 89.29% accuracy, with ROC-AUC and PR-AUC identical to those of the full model, showing that threshold selection does not alter ranking-based discrimination but does improve the final classification decision. This distinction is important: ROC-AUC and PR-AUC evaluate probability ranking across thresholds, while accuracy, sensitivity, specificity, and F1-score depend on the selected operating threshold.

Table 9 reports bootstrap confidence intervals for the proposed model. Bootstrap resampling provides a practical estimate of metric variability across repeated samples from the available test cohort. The reported mean accuracy is 91.04%, with a 95% confidence interval from 86.31 to 95.24%. The confidence interval for F1-score ranges from 87.01 to 95.48%, while ROC-AUC ranges from 89.64 to 97.51%. These intervals indicate that the proposed model remains stable under resampling variation, although the interval width also reflects the limited size of the independent test set.

Calibration was further assessed on the independent test set. The model achieved a Brier score of 0.0908, a calibration intercept of 0.3591, a calibration slope of 1.2717, and a calibration-in-the-large value of 0.3066. The observed event rate was 0.5417 compared with a mean predicted probability of 0.5071, indicating mild overall underprediction. As

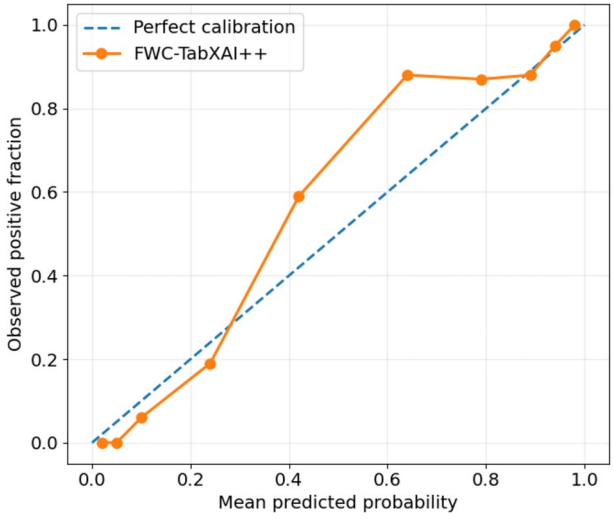


Fig. 10 Calibration plot of FWC-TabXAI++ on the independent test set

Table 10 Uncertainty-based prediction grouping

Prediction group	Number of samples	Percentage (%)	Clinical interpretation
Confident class 1	70	41.67	High-confidence heart-disease prediction
Confident class 0	62	36.90	High-confidence non-heart-disease prediction
Uncertain prediction	36	21.43	Requires clinical review or additional patient information

shown in Fig. 10, the reliability curve remained reasonably close to the ideal diagonal, with some deviation in the intermediate probability range. Probability blending in FWC-TabXAI++ represents ensemble aggregation and was not considered a separate calibration procedure.

Table 10 summarises uncertainty-based prediction grouping using the proposed model's probability output. The model assigned 70 samples to the confident heart-disease group, 62 samples to the confident non-heart-disease group, and 36 samples to the uncertain prediction group. Thus, 21.43% of test samples were flagged as uncertain. In a decision-support setting, uncertain cases can be routed for physician review, additional ECG interpretation, laboratory testing, or repeat clinical assessment. This helps reduce the risk of overreliance on automated predictions. It demonstrates how the proposed framework can be integrated into a triage workflow where high-confidence predictions support rapid decision-making and uncertain predictions trigger further evaluation.

4.8 Explainability analysis

The evaluation stage includes standard classification metrics, diagnostic metrics, ROC and precision-recall analysis, ablation testing, confidence intervals, uncertainty grouping, and explainability. The figure therefore indicates that the study is not limited to accuracy comparisons but also evaluates discrimination, reliability, clinical operating behaviour, and interpretability.

Figure 11 shows the contribution weights assigned to component learners in the FWC-TabXAI++ ensemble. The figure makes the ensemble mechanism more transparent by

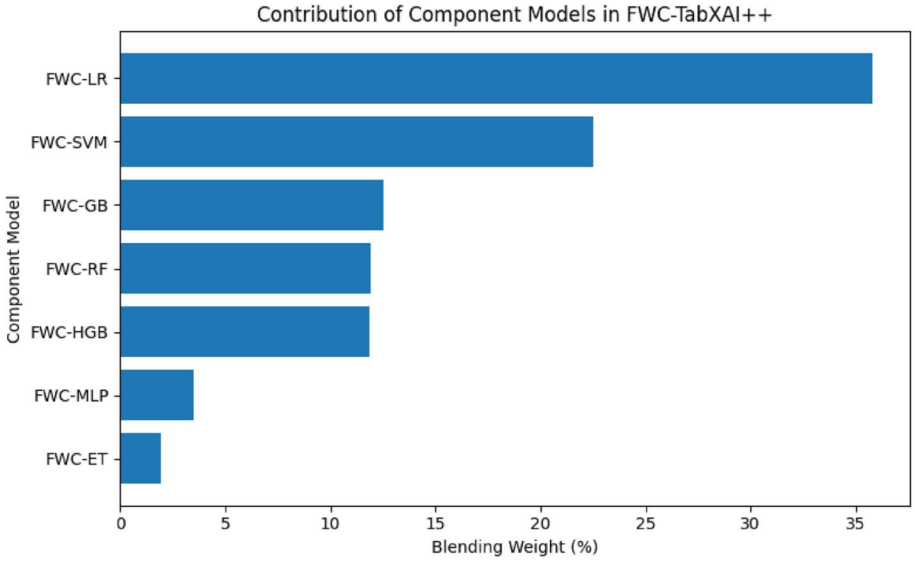


Fig. 11 Contribution weights of component models in FWC-TabXAI++

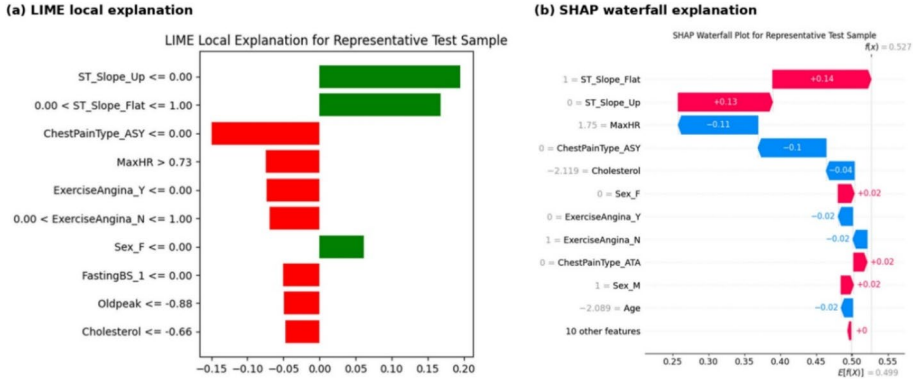


Fig. 12 Local explainability for a representative test sample using LIME and SHAP

identifying how much each learner contributes to the final probability estimate. This is important because ensemble models are often criticised for reduced interpretability. The validation-derived weights indicate that the ensemble is not a simple majority vote. Instead, it is a weighted probability aggregation in which stronger or more complementary learners receive greater influence. The inclusion of multiple models enables the framework to integrate diverse inductive biases, including linear separation, nonlinear margins, tree-based partitioning, and neural feature interactions.

Figure 12 combines LIME and SHAP waterfall explanations for a representative test sample. The left panel provides a LIME local explanation, where green bars indicate feature conditions that increase the probability of the predicted class, and red bars indicate conditions that decrease it. The right panel provides a SHAP waterfall explanation, which decomposes the prediction from the expected model output to the final patient-specific output. For the selected sample, ST_Slope_Flat and ST_Slope_Up are dominant positive contributors in the SHAP explanation, while MaxHR, ChestPainType_ASY, and cholesterol-related values reduce the prediction. The final SHAP model output is approximately $f(x) = 0.527$, close to the expected value $E[f(X)] = 0.499$, indicating a borderline

prediction. The LIME explanation provides a complementary local approximation of the decision boundary around the selected patient. It identifies rule-like feature intervals such as ST_Slope_Up ≤ 0, ST_Slope_Flat > 0, MaxHR > 0.73, and cholesterol ≤ −0.66. Presenting LIME and SHAP together strengthens the explainability analysis by showing that the model's decision can be interpreted through both local surrogate rules and additive feature attribution. SHAP and LIME quantify feature contributions to model predictions and should not be interpreted as evidence of causal relationships or clinical risk causation.

5 Conclusion and future scope

In this study, we developed FWC-TabXAI++, a feature-weighted, calibrated tabular explainable AI framework for heart disease prediction. The framework integrates clinically informed preprocessing, second-order interaction expansion, mutual-information-based feature selection, a heterogeneous ensemble of multiple classifiers, and validation-guided probability blending and threshold optimisation. The resulting model achieved balanced diagnostic performance with 91.07% accuracy, 92.22% precision, 91.21% recall, 90.91% specificity, and robust discrimination metrics (94.01% ROC-AUC and 93.24% PR-AUC). The model also showed favourable probability calibration, with a Brier score of 0.0908 and reliable bootstrap confidence intervals, indicating model stability and statistical robustness. Importantly, the integrated use of SHAP and LIME provided both global and patient-specific explanations, highlighting clinically meaningful predictors such as ST-segment slope, exercise-induced angina, and serum cholesterol, thereby enhancing the model's transparency and acceptability for clinical decision support.

Although the results are promising, the study is based on a single public dataset and should therefore be interpreted with caution regarding broader clinical generalizability. External validation on independent multi-centre cohorts is needed to confirm robustness across different populations, acquisition settings, and clinical environments. Future work should therefore focus on external validation across diverse populations and healthcare settings, integration of additional biomarkers and multi-modal data, and prospective deployment in clinical workflows. In addition, future research should explore adaptive learning mechanisms to continuously update the model with new data and investigate fairness and bias-mitigation strategies to ensure equitable predictions across patient subgroups. By addressing these avenues, FWC-TabXAI++ can evolve into a more generalisable, trustworthy, and clinically actionable tool for cardiovascular risk prediction and management.

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Author contributions
M.A. and E.A. conceptualized the study and designed the methodology. S.J. and R.B. performed data preprocessing, feature engineering, and experimental implementation. M.R.H. and P.D.R. conducted model evaluation, statistical analysis, and interpretation of results. M.T.I. and M.T.R.K. prepared the figures, tables, and explainability analysis. P.M. supervised the work, reviewed the technical content, and edited the final manuscript. All authors reviewed and approved the final version of the manuscript.

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Data availability
The dataset used and analysed during the current study is publicly available from the Kaggle repository and is based on the UCI Heart Disease dataset: https://www.kaggle.com/datasets/amirmahdiabbootalebi/heart-disease.

Declarations

Ethics approval and consent to participate
Not applicable. This study involved secondary analysis of a publicly available, anonymised dataset and did not involve direct recruitment of human participants, intervention with participants, or access to personally identifiable information.

Consent for publication
Not applicable. The manuscript does not contain identifiable personal information, individual clinical details, images, or videos requiring consent for publication.

Competing interests
The authors declare that they have no competing interests.

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